Ke Huang

CONTACT INFORMATION

UMass Lowell, Southwick Hall 350J,
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Google Scholar

RESEARCH INTERESTS

- **Statistical and AI applications in health**: develop AI and statistical models to analyze health data (e.g., human brain connectivity, Electroencephalogram (EEG), Electrocardiogram (ECG)), providing insights and advancements.
- **Statistical learning theory of Neural Networks**: provide theoretical foundations for deep learning, including approximation theory and statistical estimation theory of neural networks; explore methodologies to overcome high-dimensional challenges.

EDUCATION

University of California Riverside, Riverside, CA

Ph.D. in Applied Statistics (GPA: 3.86/4) Sep 2024
Advisor: Dr. Shujie Ma
Thesis: Statistical Learning and Applications in Sparse Neural Networks

Texas A&M University, College Station, TX

M.S. in Statistics (GPA: 3.9/4) May 2019

Jiangsu Normal University, Xuzhou, China

B.S. Applied Statistics (GPA: 3.8/4) May 2017

PROFESSIONAL EXPERIENCE

University of Massachusetts, Lowell
Assistant professor, Department of Mathematics and Statistics Sep 2026 – now

Weill Cornell Medicine & **Cornell University**
Research associate, Department of Radiology Sep 2024 – Sep 2026
Advisor: Dr. Amy Kuceyeski

- Developed statistical and AI models to analyze neuroscience data.
- Mentored undergraduate students who majored in computer science on research applying AI and statistical models to brain-related data analysis.

Neurocrine Biosciences
Research Intern Jun 2023–Aug 2023

- Developed AI/ML models to predict and characterize drug absorption and metabolic profiles; selected the significant molecular descriptors.
- Deployed a drug profile prediction webpage on AWS for enterprise use with accuracy on par with industry-standard Predictor.

Publications

Huang K., Jamison K., Miolane N., Jacobs E., Kuceyesky, A. (2026) The progression of sex differences in brain networks across the lifespan. *(Covered in Nature: https://www.nature.com/articles/d41586-026-00349-7).*

Huang K., Ma S. (2026) On the Learning Theory of Deep Convolutional Neural Network Estimators Obtained via Empirical Risk Minimization. *Under revision at Journal of Machine Learning Research.*

Huang K., Liu M., Ma S. (2025) Nearly Optimal Learning using Sparse Deep ReLU Networks in Regularized Empirical Risk Minimization with Lipschitz Loss. *Neural Computation 37(4), 815-870.*

Cheng X., **Huang K.**, Ma S. (2025). Generalization and Risk Bounds for Recurrent Neural Networks. *Neurocomputing, 616, 128825.*

Cheng X., **Huang K.**, Zou Y., Ma S. (2024). EGAN: A GAN-enhanced Ensemble Deep Learning Model for Imbalanced Classification of Sleep Stages. *Biomedical Signal Processing and Control, 92, 106020.*

Han J., Zhang B., Zhang X., **Huang K.**, Fang V., Xu X. (2023). Associations between Occurrence of Birth Defects and Hydraulic Fracturing Activities in Barnett Shale Region, Texas. *Heliyon*, 9(4).

Zhang X., **Huang K.**, Saad F., Haider K. S., Haider A., Xu, X. (2021). Testosterone Therapy Reduces Cardiovascular Risk Among Hypogonadal Men: A Prospective Cohort Study in Germany. *Clinical Research and Therapeutics, 2(1), 64-72.*

Baek J., **Huang K.**, Conner L., Tapangan N., Xu X., Carrillo G. (2019). Effects of the Home-based Educational Intervention on Health Outcomes among Primarily Hispanic Children with Asthma: a Quasi-experimental Study. *BMC Public Health, 19(1), 1-8.*

Teaching

University of Massachusetts, Lowell
Instructor
MATH 3860 Probability and Statistics I — Fall 2026

University of California, Riverside

Teaching Assistant

STAT 183 Statistical Consulting	Spring 2022
STAT 171 General Statistical Models	Winter 2022, 23, 24
STAT 155 Probability and Statistics for Science and Engineering	Fall 2021, 22
STAT 107 Introduction to Statistical Computing With R	Winter 2024
STAT 011 Introduction to Statistical Modeling	Fall 2020, 22; Winter 2021, 23
STAT 010 Introduction to Statistics	Fall 2020, 22; Summer 2021; Winter 2021, 22
STAT 008 Statistics for Business	Spring 2021; Summer 2022; Fall 2023

Mentoring

Cornell University

Katerina Popova, Undergraduate in Computer Science	Sep 2024 – May 2026
Aditya Lyer, Undergraduate in Computer Science	Sep 2024 – July 2025

Presentations

2026 Organization for Human Brain Mapping, Bordeaux, France. (Jun 2026)

Cornell Neuro Club, Cornell University, Ithaca. (March 2026)

2025 Organization for Human Brain Mapping, Brisbane, Australia. (Jun 2025)

The 14th International Conference of the ERCIM WG on Computational and Methodological Statistics, Online. (Dec 2021)

Awards

Dissertation Completion Fellowship Award, UCR	2024
Dissertation Year Fellowship Award, UCR	2023
Morris J. Garber Award, UCR	2021
Graduate Enhanced Statistic Scholarship, TAMU	2018

Service

Reviewer: Statistical Theory and Related Fields	2025
Reviewer: Nature Communications	2026